Project Report: MT3444 Combinatorial Optimization

Name and Roll Number: Vadgama Devanshu Diljeet, 20221088

1 Problem Description and Literature Review

Problem Name: Multicommodity Flow Problem

Input: A directed graph $G = (V, E)$ with edge capacities $c_e \geq 0$ for each $e \in E$; an integer k (number of commodities); for each commodity $i = 1, \ldots, k$, a source $s_i \in V$, a sink $t_i \in V$, and a demand $d_i \geq 0$.
Output: YES if there exists flows $\{f_i(e)\}$ such that each commodity i routes exactly d_i units from s_i to t_i and the total flow on each edge does not exceed c_e; NO otherwise.

Toy Example

Let $V = \{A, B, C\}$ and edges:

$$A \to B \ (c = 3), \quad B \to C \ (c = 5), \quad A \to C \ (c = 2).$$

Commodities:

- Commodity 1: $s_1 = A$, $t_1 = C$, $d_1 = 3$
- Commodity 2: $s_2 = B$, $t_2 = C$, $d_2 = 1$

A feasible routing exists by splitting commodity 1 across paths $A \to B \to C$ and $A \to C$ while routing commodity 2 via $B \to C$.

Literature Review

Multicommodity flow is a central topic in combinatorial optimization, and several foundational papers have shaped its theoretical development. Here we review three of the most influential works.

1. Leighton and Rao (1999)
Multicommodity Max-Flow Min-Cut Theorems and Their Use in Designing Approximation Algorithms, J. ACM

Leighton and Rao established that the **maximum concurrent multicommodity flow** is within a factor $\Theta(\log n)$ of the **sparsest cut** in general graphs. This result quantified the

flow–cut gap and proved that it is inherently logarithmic. Their work introduced new techniques connecting multicommodity flows to approximation algorithms for graph partitioning problems, providing the first polylogarithmic approximation bounds. It laid the foundation for understanding structural limitations and performance guarantees in multicommodity flow analysis.

2. Garg, Vazirani and Yannakakis (1996)
Approximate Max-Flow Min-(Multi)Cut Theorems and Their Applications, SIAM J. Comput.

This paper developed an **approximation algorithm** for multicommodity flow using a **primal-dual framework**. The authors proved an $O(\log n)$-approximation guarantee for related multicut problems and showed how linear programming duality could guide efficient algorithm design. This work was critical in providing an **algorithmic perspective** on multicommodity flows, demonstrating that although exact solutions may be difficult, near-optimal flows can be efficiently computed.

3. Linial, London and Rabinovich (1995)
The Geometry of Graphs and Some of its Algorithmic Applications, Combinatorica

Linial, London, and Rabinovich connected the **dual of the multicommodity flow LP** to **embedding graph metrics into ℓ_1 space**. They showed that any n-point metric can be embedded with distortion $O(\log n)$, explaining geometrically why the flow–cut gap is logarithmic. This paper introduced a **geometric viewpoint**, which has been fundamental in understanding the structural properties of multicommodity flows and their dual formulations.

Collectively, these works provide the **theoretical foundation** of multicommodity flow analysis: Leighton and Rao quantified the structural limits, Garg et al. provided algorithmic frameworks, and Linial et al. gave the geometric interpretation of the flow–cut gap. Together, they continue to influence both theory and approximation techniques in this field.

2 Linear Programming Formulation

Decision Variables

For each commodity i and edge e:

$$f_i(e) \geq 0$$

$f_i(e)$ represents the amount of flow of commodity i sent through edge e. So the set :

$$\{f_i(e) \in \mathbb{R}_{\geq 0} \mid i \in \{1, \ldots, k\},\ e \in E\}$$

represents all the decision variables.

Objective Function

This problem does not include an objective function, since we are only interested in determining whether a feasible solution exists and are not attempting to optimize any function of the form $c^T x$.

If desired, the formulation can be converted into a standard linear program by introducing a trivial objective function, for example:

$$\min\ 0.$$

Constraints

(1) Flow Conservation Constraints For each commodity i and vertex v:

$$\sum_{e=(u,v)} f_i(e) - \sum_{e=(v,w)} f_i(e) = \begin{cases} d_i & v = t_i \\ -d_i & v = s_i \\ 0 & \text{otherwise} \end{cases}$$

Explanation: Ensures that each commodity behaves like a valid flow from its source to sink.

(2) Capacity Constraints For each edge e:

$$\sum_{i=1}^{k} f_i(e) \leq c_e$$

Explanation: Total flow on any edge cannot exceed its capacity. This is the coupling constraint between commodities.

[0]In multicommodity flow, the edge capacity constraints are called *coupling constraints* because they link the flows of all commodities through the same edge, preventing each commodity from being optimized independently.

(3) Nonnegativity

$$f_i(e) \geq 0 \quad \forall\, e \in E \textit{ and } i \in [k]$$

Feasible Region Discussion

The feasible region is a convex polyhedron in $\mathbb{R}^{k|E|}$.

- It is non-empty if and only if all demands can be routed simultaneously.
- Since $0 \leq f_i(e) \leq c_e$, the region is bounded.
- Therefore, if feasible, the LP describes a bounded polytope.

For the toy example above, the feasible region is non-empty and bounded.

3 Dual Linear Program

Primal LP in Standard Form

Let $G = (V, E)$ be a directed graph with $|V| = n$ vertices and $|E| = m$ edges. There are k commodities. For commodity i, let s_i be the source, t_i the sink, and d_i the demand.

Decision variables. Let $f_i(e)$ denote the flow of commodity i on edge e. Stack all flows into the vector

$$f = \begin{pmatrix} f^1 \\ f^2 \\ \vdots \\ f^k \end{pmatrix} \in \mathbb{R}^{mk}, \quad f^i = \begin{pmatrix} f_i(e_1) \\ \vdots \\ f_i(e_m) \end{pmatrix} \in \mathbb{R}^m.$$

Node–edge incidence matrix. Let $B \in \mathbb{R}^{n \times m}$ be the incidence matrix of G:

$$B_{v,e} = \begin{cases} +1 & \text{if } e = (u, v), \\ -1 & \text{if } e = (v, w), \\ 0 & \text{otherwise.} \end{cases}$$

Primal LP constraints.

- **Flow conservation (for each commodity i):**

$$Bf^i = b^i, \quad b^i_v = \begin{cases} d_i & v = s_i, \\ -d_i & v = t_i, \\ 0 & \text{otherwise.} \end{cases}$$

where , $\sum_{e \in \delta^+(v)} f_i(e) - \sum_{e \in \delta^-(v)} f_i(e) = b^i_v$

- **Edge capacity:**

$$\sum_{i=1}^{k} f_i(e) \le c_e, \quad \forall e \in E.$$

- **Nonnegativity:**

$$f_i(e) \ge 0, \quad \forall i, \forall e.$$

Matrix form (standard LP). Stacking flow conservation constraints for all commodities:

$$A_{\text{flow}} = \begin{pmatrix} B & 0 & \cdots & 0 \\ 0 & B & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & B \end{pmatrix} \in \mathbb{R}^{nk \times mk}, \quad b_{\text{flow}} = \begin{pmatrix} b^1 \\ \vdots \\ b^k \end{pmatrix}.$$

Stacking capacity constraints:

$$A_{\text{cap}} = \begin{pmatrix} I & I & \cdots & I \end{pmatrix} \in \mathbb{R}^{m \times mk}, \quad b_{\text{cap}} = c \in \mathbb{R}^m.$$

Full standard form:

$$\text{find } f \ge 0 \text{ s.t. } \begin{cases} A_{\text{flow}} f = b_{\text{flow}}, \\ A_{\text{cap}} f \le b_{\text{cap}}. \end{cases}$$

Dual Linear Program

Dual Variables.

- For each commodity i, introduce a vector $\pi^i \in \mathbb{R}^n$ corresponding to the flow conservation constraints $A_{\text{flow}} f = b_{\text{flow}}$. Each π^i can be interpreted as a set of *vertex potentials* for commodity i, representing the "value" of having flow at each node.
- Introduce a vector $y \in \mathbb{R}^m_{\ge 0}$ corresponding to the edge capacity constraints $A_{\text{cap}} f \le b_{\text{cap}}$. Each y_e represents the *shadow price* of edge e, i.e., how tight or "expensive" it is to use that edge.

Dual LP (matrix form). The dual linear program is given by:

$$\max\ b_{\text{flow}}^T \pi - c^T y \quad \text{s.t.} \quad A_{\text{flow}}^T \pi + A_{\text{cap}}^T y \le 0, \quad y \ge 0.$$

Expanded dual constraints. Writing the constraints for each commodity i and each edge $e = (u, v)$ explicitly, we have:

$$\pi_u^i - \pi_v^i + y_e \le 0, \quad \forall i, \forall e = (u, v) \in E.$$

This shows that the potential difference across an edge, plus its shadow price, cannot exceed zero.

Expanded dual objective. The dual objective can also be written as:

$$\max \sum_{i=1}^{k} d_i(\pi^i_{s_i} - \pi^i_{t_i}) - \sum_{e \in E} c_e y_e$$

where:

- The first term $\sum_i d_i(\pi^i_{s_i} - \pi^i_{t_i})$ represents the total "benefit" of sending the required demand from sources to sinks for all commodities.
- The second term $\sum_{e \in E} c_e y_e$ represents the total "cost" of edge usage due to capacity limitations.

Interpretation.

- $\pi^i_u - \pi^i_v$ measures the "cost" or potential difference of moving one unit of commodity i from node u to node v.
- y_e measures how tight the capacity on edge e is — its marginal value: how much increasing capacity would improve feasibility.
- The dual objective maximizes the net benefit of sending all commodity demands while accounting for edge capacity limitations.
- Even though the primal LP is a feasibility problem with objective 0, the dual provides a meaningful maximization problem that reflects the feasibility conditions in terms of vertex potentials and shadow prices.

References

[1] D. P. Bertsekas, *Network Optimization: Continuous and Discrete Models*, Athena Scientific, 1998.

[2] T. Leighton and S. Rao, *Multicommodity Max-Flow Min-Cut Theorems and Their Use in Designing Approximation Algorithms*, Journal of the ACM, 1999.

[3] N. Garg, V. V. Vazirani, and M. Yannakakis, *Approximate Max-Flow Min-(Multi)Cut Theorems and Their Applications*, SIAM Journal on Computing, 1996.

[4] N. Linial, E. London, and Y. Rabinovich, *The Geometry of Graphs and Some of its Algorithmic Applications*, Combinatorica, 1995.

[5] OpenAI, *ChatGPT*, https://chat.openai.com/, Accessed: 2026.